c-only slope law: process noise q_{55} of the constant c

1 $\mathbf{F}_e$ and $\mathbf{H}$

$$\mathbf{e}[k] = \begin{bmatrix} e_{\delta\bar{w}_1}[k] \\ e_{\delta\bar{w}_2}[k] \\ e_{\delta\bar{w}_3}[k] \\ e_{\bar{a}_w}[k] \\ e_c[k] \end{bmatrix}, \qquad \begin{aligned} \hat{h}[k] &= \bar{w}_d[k] - \delta\hat{\bar{w}}_3[k] - \bar{w}_s \\ \bar{a}'[k] &= \frac{1-\hat{c}[k]}{\hat{h}[k]^2} \\ M[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k] \\ \nabla_d\bar{w}_d[k] &= \bar{w}_d[k] - \bar{w}_d[k-d] \end{aligned}$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw}[k] & 0 \\ 0 & 0 & \bar{a}'[k]\,(1-\lambda_c) + \dfrac{2\,\bar{a}'[k]}{\hat{h}[k]}\,M[k] & 1+\bar{a}'[k]\,F_{dw}[k] & -\dfrac{M[k]}{\hat{h}[k]^2} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{H}[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\dfrac{2\,\bar{a}'[k]}{\hat{h}[k]}\,\nabla_d\bar{w}_d[k] & 1 & \dfrac{\nabla_d\bar{w}_d[k]}{\hat{h}[k]^2} \end{bmatrix}$$

Where c appears in $\mathbf{F}_e$ and $\mathbf{H}$

$$\mathbf{F}_e[k](5,:) = \begin{bmatrix} 0, & 0, & 0, & 0, & 1 \end{bmatrix}$$

$$f_{e,45}[k] = -\frac{M[k]}{\hat{h}[k]^2}$$

$$h_{15}[k] = 0$$

$$h_{25}[k] = \frac{\nabla_d\bar{w}_d[k]}{\hat{h}[k]^2}$$

At hold ($\Delta\bar{w}_d = 0$ for d steps)

$$f_{e,45}[k] = -\frac{(1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]}{\hat{h}[k]^2}$$

$$h_{25}[k] = 0$$

2 Kalman filter

$$\mathbf{L}[k] = \mathbf{P}^f[k]\,\mathbf{H}^T[k]\big(\mathbf{H}[k]\,\mathbf{P}^f[k]\,\mathbf{H}^T[k] + \mathbf{R}[k]\big)^{-1}$$

$$\mathbf{P}[k] = \big(\mathbf{I} - \mathbf{L}[k]\,\mathbf{H}[k]\big)\,\mathbf{P}^f[k]$$

$$\mathbf{P}^f[k+1] = \mathbf{F}_e[k]\,\mathbf{P}[k]\,\mathbf{F}_e^T[k] + \mathbf{Q}[k]$$

3 Variance of c in the filter

Gain on c (row 5 of $\mathbf{L}[k]$)

$$\big[\,\ell_{51}[k],\ \ell_{52}[k]\,\big] = \Big[\,p^f_{51}[k],\quad h_{23}[k]\,p^f_{53}[k] + p^f_{54}[k] + h_{25}[k]\,p^f_{55}[k]\,\Big]\big(\mathbf{H}[k]\,\mathbf{P}^f[k]\,\mathbf{H}^T[k] + \mathbf{R}[k]\big)^{-1}$$

Update

$$\hat{c}[k+1] = \hat{c}[k] + \ell_{51}[k]\,e_{y_1}[k] + \ell_{52}[k]\,e_{y_2}[k]$$
$$p_{55}[k] = p^f_{55}[k] - \ell_{51}[k]\,p^f_{15}[k] - \ell_{52}[k]\,\Big(h_{23}[k]\,p^f_{35}[k] + p^f_{45}[k] + h_{25}[k]\,p^f_{55}[k]\Big)$$

Information gain on c at step k (computed from the update)

$$I[k] = \frac{1}{p_{55}[k]} - \frac{1}{p^f_{55}[k]} \;\geq\; 0$$
$$p^f_{55}[k] - p_{55}[k] = p^f_{55}[k]\,p_{55}[k]\;I[k]$$

Prediction ($\mathbf{F}_e[k](5,:) = [\,0,\ 0,\ 0,\ 0,\ 1\,]$)

$$p^f_{55}[k+1] = p_{55}[k] + q_{55}[k], \qquad p^f_{55}[0] = P_0$$

4 Choice of q_{55}: constant variance $p^f_{55} = p_{\text{fix}}$

Condition

$$p^f_{55}[k+1] = p^f_{55}[k] = p_{\text{fix}}$$

Prediction and information gain under the condition

$$q_{55}[k] = p_{\text{fix}} - p_{55}[k]$$
$$\frac{1}{p_{55}[k]} = \frac{1}{p_{\text{fix}}} + I[k]$$
$$q_{55}[k] = \frac{p^2_{\text{fix}}\,I[k]}{1 + p_{\text{fix}}\,I[k]}$$

$p_{\text{fix}}\,I[k] \ll 1$

$$\boxed{q_{55}[k] = p^2_{\text{fix}}\,I[k]}$$

5 Meaning of $I[k]$ (scalar case: c alone, measurement y_2 only)

Output error caused by e_c

$$e_{y_2}[k] = h_{25}[k]\, e_c[k] + n_a[k], \qquad \mathrm{Var}\big(n_a[k]\big) = r_2[k]$$

c read from this step alone

$$\frac{e_{y_2}[k]}{h_{25}[k]} = e_c[k] + \frac{n_a[k]}{h_{25}[k]}$$

$$\mathrm{Var}\Big(\frac{n_a[k]}{h_{25}[k]}\Big) = \frac{r_2[k]}{h_{25}[k]^2} = \frac{1}{I[k]}$$

Update = combination of $\hat{c}$ (variance p^f_{55}) and this reading of c (variance $1/I$)

$$\frac{1}{p_{55}[k]} = \frac{1}{p^f_{55}[k]} + I[k]$$

$$\ell_{52}[k]\, h_{25}[k] = \frac{p^f_{55}[k]\, I[k]}{1 + p^f_{55}[k]\, I[k]} \;\simeq\; p^f_{55}[k]\, I[k]$$

6 $I[k]$ along ramp_tri ($\sqrt{P_0} = 0.1$, $\sqrt{p_{\mathrm{fix}}} = 0.15$, 10 seeds)

